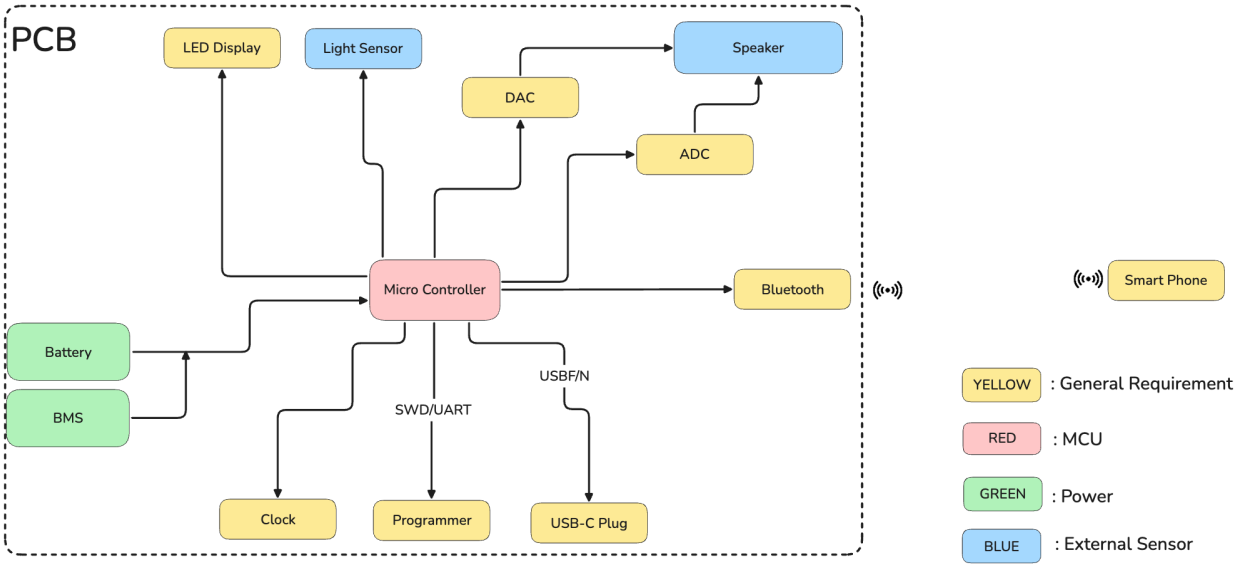
Javier Aldana
9/7/2026
MM Week 3

Hours worked: 6 Hours

3 hours worked during scheduled lab hours discussing team contract, what project to pursue, and what

3 hours worked on individually coming up with requirements (mainly hardware) and a rough system diagram of what the multi functional panel should do.



Requirements - Wall Mounted Multi Use Panel

Physical

- **Shall** be big enough to see well from across the room (12” x 8”)?
- **Shall** have an electrically controlled display
- **Shall** be able to notify users of certain external environment conditions.
- **Shall** be able to function as a wall clock.
- **Should** have a speaker to sound an alarm.
- **Should** have a light detection to lower the brightness as it gets darker.
- **Shall** have wall mounts to be relatively easy to install on a wall.

Electrical

- **Shall** have an industry standard MCU (STM32, ESP)
    - **Should** have Bluetooth (if not some module can be added)
    - **Shall** be programmed from laptop
    - **Should** have SWD, USB, UART comm protocols
    - **Should** have onboard ADC
- **Shall** be able to work off of batteries.
- **Shall** filter any analog signal it takes in.
- **Should** be able to receive power and be programmed via usb cable for testing purposes.
- **Shall** have a microphone or method for listening to dB.
    - **Should** be noise resistant and avoid false positives.
- **Should** have a charge that lasts longer than a week.
- **Should** have a DAC to create a speaker sound.
- **Should** have a real time clock function, ideally innate to the PCB
- **Should** use minimal power when not directly being used but not be turned off.
- **Shall** have a battery management system in the case of a battery being added.

Software

- **Shall** be able to set multiple functions via smartphone.
- **Should** be able to be programmed via smartphone.